

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

NIHARIKA PREEJESH (192524506)

Aim

To write a Python program to implement the **Tic-Tac-Toe game** using a 3×3 game board.

Objective

- To implement a two-player game using Python.
- To represent the game board using a list.
- To check winning conditions.
- To handle player turns and invalid moves.

Algorithm

1. Start the program.
2. Create a 3×3 board with positions 1–9.
3. Assign X to Player 1 and O to Player 2.
4. Display the board.
5. Ask the current player to select a position.
6. Check whether the selected position is empty.
7. Place the player's symbol in that position.
8. Check all rows, columns, and diagonals for a winner.
9. If a player wins, display the winner and stop.
10. If all positions are filled without a winner, declare a draw.
11. Otherwise, switch to the other player.

12. Repeat until the game ends.

Python Code

Tic-Tac-Toe Game

```
board = [" " for i in range(9)]
```

Display the board

```
def display_board():
```

```
    print()
```

```
    print(" " + board[0] + " | " + board[1] + " | " + board[2])
```

```
    print("---+---+---")
```

```
    print(" " + board[3] + " | " + board[4] + " | " + board[5])
```

```
    print("---+---+---")
```

```
    print(" " + board[6] + " | " + board[7] + " | " + board[8])
```

```
    print()
```

Check whether a player has won

```
def check_winner(player):
```

```
    winning_positions = [
```

```
        (0, 1, 2),
```

```
        (3, 4, 5),
```

```
(6, 7, 8),  
(0, 3, 6),  
(1, 4, 7),  
(2, 5, 8),  
(0, 4, 8),  
(2, 4, 6)  
]
```

```
for a, b, c in winning_positions:  
    if board[a] == player and board[b] == player and board[c] ==  
player:  
        return True  
  
return False
```

```
# Check whether board is full
```

```
def board_full():  
    return " " not in board
```

```
# Main game
```

```
print("TIC-TAC-TOE")
```

```
print("Positions are:")
```

```
print("1 | 2 | 3")
print("---+---+---")
print("4 | 5 | 6")
print("---+---+---")
print("7 | 8 | 9")
```

```
current_player = "X"
```

```
while True:
```

```
    display_board()
```

```
    print("Player", current_player, "turn")
```

```
    try:
```

```
        position = int(input("Enter position (1-9): "))
```

```
        if position < 1 or position > 9:
```

```
            print("Invalid position! Enter 1-9.")
```

```
            continue
```

```
        index = position - 1
```

```
        if board[index] != " ":
```

```
    print("Position already occupied!")  
    continue
```

```
    board[index] = current_player
```

```
except ValueError:
```

```
    print("Please enter a number.")  
    continue
```

```
# Check winner
```

```
if check_winner(current_player):
```

```
    display_board()  
    print("Player", current_player, "wins!")  
    break
```

```
# Check draw
```

```
if board_full():
```

```
    display_board()  
    print("Game Draw!")  
    break
```

```
# Change player
```

```
if current_player == "X":
```

```
    current_player = "O"
```

else:

current_player = "X"

Sample Input / Output

Suppose Player X enters:

1

Player O:

2

Player X:

4

Player O:

5

Player X:

7

Output

X | O |

---+---+---

X | O |

---+---+---

X | |

Player X wins!

The winning combination is:

1

4

7

which is the **first column**.

Example Game

TIC-TAC-TOE

Positions are:

1 | 2 | 3

---+---+---

4 | 5 | 6

---+---+---

7 | 8 | 9

Player X turn

Enter position (1-9): 1

Player O turn

Enter position (1-9): 5

Player X turn

Enter position (1-9): 2

Player O turn

Enter position (1-9): 8

Player X turn

Enter position (1-9): 3

Player X wins!

Result

Thus, the **Tic-Tac-Toe game was successfully implemented in Python** using a 3×3 board, player turns, winning-condition checking, invalid-input handling, and draw detection.